

that all families from Tambolia keep their houses dirty. Again by induction, another native resident may develop the rule that all *young* families from Tambolia keep their houses dirty. The two rules can be formulated as the following prules:

If the family is from Tambolia,
then it keeps its house dirty.

If the family is from Tambolia,
and the family is young,
then it keeps its house dirty.

The first prule will fire for all Tambolian families, whereas the second prule will fire only for young Tambolian families, which are a proper subset of all Tambolian families. So an elderly Tambolian family will keep its house dirty according to the first prule, but not according to the second.

The two native residents have learnt differently from the same training set of the three young Tambolian families. We may say that both these prules typify bigotry. Thus, one can learn bigotry by using induction, but then unlearn it, again by induction. Suppose another young Tambolian family moves into the neighbourhood and keeps its house clean. One would then realize that both the earlier prules need to be modified because it is evident that not all young Tambolian families keep their houses dirty. Learning something by induction often consists of iteratively refining what we had learnt earlier, which could sometimes also mean unlearning what has been learnt earlier.

Bigotry can be learned by induction, but that does not mean induction is to be discouraged. On the contrary, it can be encouraged: doctors continually improve their inductive learning over the years, as they see more and more patients (similar to adding more patterns to their training set), and that leads to a continuous improvement in their skill of treating patients. Just as a surgeon is expected to use his knife to remove a tumour, not slit a throat, induction is expected to be put to good use.

1.3 A Professor-Student Example

In this example, the objective is to distinguish professors from students by observing three of their attributes HABIT, EATS, and FOOTWEAR.

NAME of training pattern	Attributes			Class
	HABIT	EATS	FOOTWEAR	
T1	Gabby	Baked	Clogs	Student
T2	Gabby	Roasted	Sandals	Professor
T3	Gabby	Baked	Sandals	Student
T4	Quiet	Fried	Sandals	Professor
T5	Gabby	Fried	Clogs	Student
T6	Quiet	Baked	Sandals	Student
T7	Gabby	Fried	Sandals	Professor
T8	Quiet	Fried	Clogs	Student

FIGURE 1.1 Training set for recognizing professors and students. The possible attribute values are 'gabby' and 'quiet' for HABIT; 'baked', 'fried', and 'roasted' for EATS; and 'clogs' and 'sandals' for FOOTWEAR.

Figure 1.1 provides a training set¹ of eight patterns of professors and students. The caption of the figure lists the possible values for each of these three attributes. The figure is, in fact, a table, but all tables, diagrams, and any textual asides will be referred to as figures in this book, for that puts all of them in a common sequence, thus making it easier to search for them.

Study Fig. 1.1, and by intuition, come up with a set of prules for each of the two classes. This is similar to what a learning program does except that it does not use intuition. Now jot down justifications for the prules developed. Apply your prules to infer the classes of the four recall patterns given in Fig. 1.2. Figures 1.1 and 1.2 will be referred to at many places in the book; hence it is advised that one becomes well familiar with them.

Now you may ask seven friends to infer the classes of the recall patterns based on their intuitive study of the training set. Do not be surprised if their classifications of the recall patterns are not the same as yours. If a

¹I got the idea of this training set while reading an example on lions in P. Clark 1990, 'Machine Learning: Techniques and Recent Developments', in A.R. Mirzai, (ed.), *Artificial Intelligence: Concepts and Application in Engineering*, MIT Press, Cambridge, Massachusetts, pp. 65-93.

NAME of recall pattern	Attributes			Class
	HABIT	EATS	FOOTWEAR	
R1	Quiet	Baked	Clogs	
R2	Quiet	Roasted	Sandals	
R3	Gabby	Roasted	Clogs	
R4	Quiet	Roasted	Clogs	

FIGURE 1.2 Recall set for recognizing professors and students. The training set is given in Fig. 1.1.

person's classifications are different from yours, then what that person learnt by induction from the training set was different from what you learnt by induction from the same training set (remember how the two native residents learnt differently about the Tambolian families in Section 1.2).

1.4 A Specific-to-General Procedure

The antecedent of a prule is said to *cover* a pattern if, and only if, the attribute values of the pattern make the antecedent true. Antecedent α_1 of a prule is defined to be *more specific* than antecedent α_2 of another prule if, and only if, the patterns covered by α_1 are a proper subset of the patterns covered by α_2 . In other words, every pattern covered by α_1 is covered by α_2 , but not conversely. In the two prules given about Tambolian families in Section 1.2, you will notice that the antecedent of the second prule is more specific than the antecedent of the first prule. Antecedent α_2 is said to be *more general* than antecedent α_1 if, and only if, α_1 is more specific than α_2 . Thus, the first prule about Tambolian families is more general than the second prule.

The Specific-to-General (abbreviated as SpecToGen) procedure to generate prules for recognizing patterns of a particular class from a given training set starts with a prule whose antecedent is so specific that it covers no training patterns. The antecedent is then gradually generalized so that it covers the training patterns of the class. This procedure can then be repeated for each of the other classes of the training set. This procedure is as follows.

5

Learning Objectives

This chapter contains the following topics:

- prior, posterior, and conditional probabilities
- maximizing posterior probabilities using Bayes classification
- reducing computation assuming the class conditional independence of attributes in Naive Bayes classification
- maximum-likelihood and Bayesian estimates of probabilities
- estimating probabilities from the professor-student training set of the first chapter, and the Naive Bayes classifications of the patterns of the corresponding recall set
- classifying patterns by minimizing the risk of misclassification
- transforming a Naive Bayes classifier into a linear classifier by using binary attributes
- Bayes or Naive Bayes classifiers using probability density functions for continuous attribute values

Bayes Classification

Simplifying Bayes Classification

A Bayes classifier does not use prules to classify patterns. It estimates probabilities of occurrence of different attribute values for the different classes in a training set. It then uses these probabilities to classify recall patterns. Let

- $\bar{A}$ be an array of $M \geq 1$ attributes $A_1, A_2, \dots, A_M$ for the patterns of a training set, and
- $P(\bar{A})$ be the probability that a training pattern has attribute array $\bar{A}$, regardless of the class to which the pattern belongs, the attributes having discrete values.

Suppose the training set has patterns from $m \geq 1$ classes $C_1, C_2, \dots, C_m$. For $1 \leq k \leq m$, we define the following probabilities.

- $P(C_k)$ is the probability that a training pattern belongs to class C_k . It is also known as the *prior* probability of class C_k .
- $P(C_k|\bar{A})$ is the probability that a training pattern with attribute array $\bar{A}$ belongs to class C_k . This is also known as the *posterior* probability of C_k for a given $\bar{A}$. The attributes have discrete values.
- $P(\bar{A}|C_k)$ is the *conditional* probability that a training pattern of class C_k has attribute array $\bar{A}$, the attributes having discrete values.
- $p(\bar{A}|C_k)$ is the conditional probability density that a training pattern of class C_k has attribute array $\bar{A}$, the attributes having continuous values.

As is the convention, upper case P is being used for conditional probability when the attribute values are discrete, and lower case p is being used when the attribute values are continuous.

To classify a pattern with attribute array $\bar{A}$, we assign it to the class most probable for $\bar{A}$. This means we estimate the posterior probability $P(C_k|\bar{A})$ for each of the classes, that is, for $k = 1$ to m . We then assign the pattern to the class with the highest such probability (if two or more classes have equal highest posterior probabilities, then the pattern is rejected). It is briefly referred to as *maximizing* $P(C_k|\bar{A})$. According to the Bayes' theorem of

probability theory

$$P(C_k|\bar{A}) = \frac{P(C_k)P(\bar{A}|C_k)}{P(\bar{A})}$$

Maximizing $P(C_k|\bar{A})$ is the same as maximizing the right-hand side of the above equation. The right-hand side denominator contains $P(\bar{A})$, which remains constant for all classes. So its value need not be estimated, hence maximizing $P(C_k|\bar{A})$ is the same as maximizing

$$P(C_k)P(\bar{A}|C_k) \quad (5.1)$$

After they have been estimated from the training set, probabilities $P(C_k)$ and $P(\bar{A}|C_k)$ can be used to classify a pattern with attribute array $\bar{A}$. For most domains, estimating $P(\bar{A}|C_k)$ needs an impractically large training set, as we need to consider all possible values for the attributes A_1 to A_M . Moreover, it needs a lot of computation, which, in turn, can require a lot of time. So it is usually assumed that the attributes A_1 to A_M are class conditionally independent, which means it is often assumed that

$$P(\bar{A}|C_k) = \prod_{i=1}^M P(A_i|C_k)$$

After making the above assumption, the classifier is called the *Naive Bayes* classifier, in which to classify a pattern with attributes A_1 to A_M , we maximize

$$P(C_k) \prod_{i=1}^M P(A_i|C_k) \quad (5.2)$$

which is obtained by substituting $\prod_{i=1}^M P(A_i|C_k)$ for $P(\bar{A}|C_k)$ in Eqn (5.1).

5.2 Estimation of Probabilities

Suppose that the number of patterns in class C_k is $|C_k|$, for $1 \leq k \leq m$. Then under the conventional estimation, known as the *maximum-likelihood* estimation,

$$P(C_k) = \frac{|C_k|}{\sum_{j=1}^m |C_j|} \quad (5.3)$$

Under an alternative estimation, called the *Bayesian* estimation,

$$P(C_k) = \frac{|C_k| + 1}{m + \sum_{j=1}^m |C_j|} \quad (5.4)$$

In the Bayesian estimation, we assume the number of patterns of each class to be one more than actually present in the training set. For a typical training set with a lot of patterns (that is, $\sum_{j=1}^m |C_j|$ is a large number), both the maximum-likelihood and Bayesian estimates give approximately equal values. Moreover, for both estimates, $\sum_{k=1}^m P(C_k) = 1$, as it should be. One difference exists between the two estimations: in the maximum-likelihood estimation, $P(C_k) \geq 0$; in the Bayesian estimation, because the numerator of the above equation is always one or more, $P(C_k) > 0$. In other words, in the Bayesian estimation a probability is never equal to zero. This is useful in practice, as we shall soon see.

As an example, the professor-student training set given in Fig. 1.1 has 3 patterns belonging to the class professor, and 5 to the class student. There being two classes, the value of m is 2. Hence, from Eqn (5.3), the maximum-likelihood estimate of $P(\text{professor})$ is $3/(3+5)$, that is $3/8$; and the Bayesian estimate of $P(\text{professor})$, from Eqn (5.4), is $(3+1)/(2+3+5)$, that is $4/10$, which may be simplified to $2/5$.

To maximize $P(C_k) \prod_{i=1}^M P(A_i|C_k)$ of Eqn (5.2), the other probability estimates required are those for $\prod_{i=1}^M P(A_i|C_k)$. Let

- the possible values for attribute A_i be $v_{i_1}, v_{i_2}, \dots, v_{i_n}$, for $1 \leq i \leq M$ and
- $|C_k^{ij}|$ be the number of training patterns of class C_k for which the value of attribute A_i is v_j , for $1 \leq k \leq m$, for $1 \leq i \leq M$, and for $i_1 \leq j \leq i_n$.

Then, for $1 \leq k \leq m$, for $1 \leq i \leq M$, and for $i_1 \leq j \leq i_n$, according to the maximum-likelihood estimation

$$P[(A_i = v_j)|C_k] = \frac{|C_k^{ij}|}{|C_k|} \quad (5.5)$$

and according to the Bayesian estimation

$$P[(A_i = v_j)|C_k] = \frac{|C_k^{ij}| + 1}{i_n + |C_k|} \quad (5.6)$$

In the Bayesian estimation, we assume that the number of training patterns of class C_k for which the value of attribute A_i is v_j is one more than actually present. Owing to this, as mentioned above, in the Bayesian estimation, no probability is ever equal to zero.

To illustrate, attribute EATS of the training set given in Fig. 1.1 has 3 possible values (baked, roasted, and fried). Of the 5 student patterns, none has the value of EATS = roasted. Therefore, from Eqn (5.5), the maximum-likelihood estimate of $P[(\text{EATS} = \text{roasted}) | \text{student}]$ is 0/5, and the Bayesian estimate, from Eqn (5.6), is $(0 + 1)/(3 + 5)$, which becomes 1/8.

The maximum-likelihood and Bayesian estimates of the various probabilities from the professor-student training set given in Fig. 1.1 are shown in Fig. 5.1. These are the probability estimates said to have been learned by the classifier. The correctness of the probability estimates given in Fig. 5.1 can be checked using Eqns (5.3) to (5.6). The fractions given in the figure have not been simplified, to avoid any confusion; for instance, a probability estimate of 4/10 appears as such, and has not been simplified to 2/5. Since our training set is not large, the maximum-likelihood estimation of a probability is not as close to the Bayesian estimation as it would have been, if the training set had been large. Nevertheless, the usefulness of Bayesian estimation will become clear in the next Section 5.3.

5.3 Classifying Professor-Student Patterns

Let us classify pattern R3 of the professor-student recall set given in Fig. 1.2. The attribute values for the pattern are HABIT = gabby, EATS = roasted, and FOOTWEAR = clogs. Using the maximum-likelihood estimates from Fig. 5.1 and the Naive Bayes classifier of Eqn (5.2), the value of

$$P(\text{professor}) \times P[(\text{HABIT} = \text{gabby}) | \text{professor}] \\ \times P[(\text{EATS} = \text{roasted}) | \text{professor}] \times P[(\text{FOOTWEAR} = \text{clogs}) | \text{professor}]$$

is $3/8 \times 2/3 \times 1/3 \times 0/3 = 0$, and the value of

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$$P(\text{student}) \times P[(\text{HABIT} = \text{gabby}) | \text{student}] \\ \times P[(\text{EATS} = \text{roasted}) | \text{student}] \times P[(\text{FOOTWEAR} = \text{clogs}) | \text{student}]$$

Probability	Estimates	
	Maximum-likelihood	Bayesian
$P(\text{professor})$	3/8	4/10
$P[(\text{HABIT} = \text{gabby}) \text{professor}]$	2/3	3/5
$P[(\text{HABIT} = \text{quiet}) \text{professor}]$	1/3	2/5
$P[(\text{EATS} = \text{baked}) \text{professor}]$	0/3	1/6
$P[(\text{EATS} = \text{fried}) \text{professor}]$	2/3	3/6
$P[(\text{EATS} = \text{roasted}) \text{professor}]$	1/3	2/6
$P[(\text{FOOTWEAR} = \text{clogs}) \text{professor}]$	0/3	1/5
$P[(\text{FOOTWEAR} = \text{sandals}) \text{professor}]$	3/3	4/5
$P(\text{student})$	5/8	6/10
$P[(\text{HABIT} = \text{gabby}) \text{student}]$	3/5	4/7
$P[(\text{HABIT} = \text{quiet}) \text{student}]$	2/5	3/7
$P[(\text{EATS} = \text{baked}) \text{student}]$	3/5	4/8
$P[(\text{EATS} = \text{fried}) \text{student}]$	2/5	3/8
$P[(\text{EATS} = \text{roasted}) \text{student}]$	0/5	1/8
$P[(\text{FOOTWEAR} = \text{clogs}) \text{student}]$	3/5	4/7
$P[(\text{FOOTWEAR} = \text{sandals}) \text{student}]$	2/5	3/7

FIGURE 5.1 Maximum-likelihood and Bayesian estimates of prior and conditional probabilities from the professor-student training set given in Fig. 1.1. A few of the maximum-likelihood estimates are zero but none of the Bayesian estimates are so. The maximum-likelihood estimates of the prior probabilities $P(\text{professor})$ and $P(\text{student})$ are obtained from Eqn (5.3), and their Bayesian estimates from Eqn (5.4). The remaining are conditional probabilities: their maximum-likelihood estimates are obtained from Eqn (5.5), and the Bayesian estimates from Eqn (5.6).

is $5/8 \times 3/5 \times 0/5 \times 3/5 = 0$. The pattern is rejected because the values for both the classes are zero. For each class, one zero probability has nullified the influence of the other probabilities. This is a disadvantage of using the maximum-likelihood estimates of probabilities. Instead, using the Bayesian estimates from Fig. 5.1, the value of

$$P(\text{professor}) \times P[(\text{HABIT} = \text{gabby}) \mid \text{professor}] \\ \times P[(\text{EATS} = \text{roasted}) \mid \text{professor}] \times P[(\text{FOOTWEAR} = \text{clogs}) \mid \text{professor}]$$

is $2/5 \times 3/5 \times 1/3 \times 1/5 = 0.016$, and the value of

$$P(\text{student}) \times P[(\text{HABIT} = \text{gabby}) \mid \text{student}] \\ \times P[(\text{EATS} = \text{roasted}) \mid \text{student}] \times P[(\text{FOOTWEAR} = \text{clogs}) \mid \text{student}]$$

is $3/5 \times 4/7 \times 1/8 \times 4/7 = 0.0245$. Since the value 0.0245 obtained for student is more than 0.016 obtained for professor, recall pattern R3 is classified as student.

Thus, it is observed that pattern R3 was rejected when maximum-likelihood estimation was used, but was classified when Bayesian estimation was used. Bayesian estimates have accordingly been found to be useful in practice.

Using the Bayesian estimates of probabilities from Fig. 5.1, the Naive Bayes classifications of the four professor-student recall patterns given in Fig. 1.2 are shown in Fig. 5.2. The classifications are incidentally identical to those shown in Fig. 2.11 for the decision tree of Fig. 2.9 (the tree obtained by maximizing the ratio of information gain).

NAME of recall pattern	Attributes			CLASSIFICATION
	HABIT	EATS	FOOTWEAR	
R1	Quiet	Baked	Clogs	Student
R2	Quiet	Roasted	Sandals	Professor
R3	Gabby	Roasted	Clogs	Student
R4	Quiet	Roasted	Clogs	Student

FIGURE 5.2 Classifications of the professor-student recall patterns given in Fig. 1.2, using the Bayesian estimates of probabilities in the Naive Bayes classifier. The Bayesian estimates were taken from Fig. 5.1, and the Naive Bayes classifier of Eqn (5.2) was used.

estimation of probabilities is at times more useful in classification than the maximum-likelihood estimation. The Bayesian classifier can alternatively base its classification on minimizing the risk of misclassification. With binary attributes, the Naive Bayes classifier becomes a linear classifier. When the attribute values are continuous, the probability density functions can be used to estimate the conditional probabilities.

Exercises

1. Manually verify the probability estimates given in Fig. 5.1.
2. Obtaining the Bayesian estimates of probabilities from Fig. 5.1, manually carry out the calculations required to classify patterns R1, R2, and R4 in Fig. 5.2 by the Naive Bayes classifier using Eqn (5.2). The calculations required to classify pattern R3 are already given in the example of Section 5.3. Again, classify these patterns using the maximum-likelihood estimates of probabilities. Given $\lambda(\text{professor}) = 2$ and $\lambda(\text{student}) = 1$, reclassify the patterns by the Naive Bayes classifier minimizing risk, by using Eqn (5.7), and by using the Bayesian estimates of probabilities.
3. Obtaining the Bayesian estimates of probabilities from Fig. 5.1, manually show that the Naive Bayes classifier using Eqn (5.2) will correctly classify all the professor-student training patterns T1 to T8 given in Fig. 1.1.
4. Suppose attribute HABIT does not exist in the professor-student training set given in Fig. 1.1 and the recall set given in Fig. 1.2. After training manually on the eight training patterns, classify the four recall patterns by the Naive Bayes classifier [Eqn (5.2)], using Bayesian probability estimates. Are the classifications same as those in Fig. 5.2, in which HABIT is an attribute? If yes, why is it so? (*Hint*: See Fig. 2.9)
5. In a programming language of your choice, write a program to implement a Naive Bayes classifier using Eqn (5.7). In the training phase, it estimates the required prior and conditional probabilities from a given training set. The user can choose to have the program give either maximum-likelihood or Bayesian estimates of probabilities. In the testing phase, (a) the user provides the value $\lambda(C_i)$, the loss in misclassifying a pattern of class C_i , for each of the m classes, that

in which

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} & \cdots & \sigma_{1M} \\ \sigma_{21} & \sigma_2^2 & \sigma_{23} & \cdots & \sigma_{2M} \\ \sigma_{31} & \sigma_{32} & \sigma_3^2 & \cdots & \sigma_{3M} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sigma_{M1} & \sigma_{M2} & \sigma_{M3} & \cdots & \sigma_M^2 \end{bmatrix}$$

and

- $|\Sigma|$ is the determinant of Σ .
- σ_i^2 is the variance of A_i over the training patterns of class C_k for $1 \leq i \leq M$.
- σ_{ij} is the covariance of A_i and A_j over the training patterns of class C_k , for $1 \leq i \leq M$ and $1 \leq j \leq M$, where $i \neq j$.
- $\bar{\mu} = \bar{\mu}_1, \bar{\mu}_2, \dots, \bar{\mu}_M$, where $\bar{\mu}_i$ is the mean of A_i over the training patterns of class C_k .
- $(\bar{A} - \bar{\mu})^t$ is the transpose of $(\bar{A} - \bar{\mu})$.

5.7 Performance of Bayes Classifier

The Bayes classifier makes the optimal decision in assigning a pattern to the most probable class. In theory, a Bayes classifier performs the least misclassifications as compared to any other classifier. In practice, the Bayes classifier usually does well, but not so well as the theory says it should. This happens mainly because of inaccuracies in estimating the probabilities. Moreover, if the Naive Bayes classifier is used, then the assumption of the attributes being class conditionally independent may not be true.

Summary

Given the attribute values of a pattern, a Bayes classifier recognizes the pattern by maximizing the posterior probability over the classes. To reduce computation, the Naive Bayes classifier, in which the attributes are assumed to be class conditionally independent, is often used. The Bayesian

Let $\bar{x}$ be an array of $n \geq 1$ numbers $x_1, x_2, \dots, x_n$. Then the *mean* μ_x of the array is defined to be

$$\frac{1}{n} \sum_{i=1}^n x_i$$

The *variance* σ^2 of array $\bar{x}$ is defined to be

$$\frac{1}{n} \sum_{i=1}^n (x_i - \mu_x)^2$$

The positive square root of the variance is defined to be the *standard deviation* σ of the array.

Let there be another array $\bar{y}$ of n numbers $y_1, y_2, \dots, y_n$ numbers, and let their mean be μ_y . The *covariance* σ_{xy} of arrays $\bar{x}$ and $\bar{y}$ is defined to be

$$\frac{1}{n} \sum_{i=1}^n (x_i - \mu_x)(y_i - \mu_y)$$

FIGURE 5.3 A review of mean, variance, standard deviation, and covariance.

(or Gaussian) distribution, hence for $1 \leq i \leq M$, we have

$$p(A_i|C_k) = \frac{1}{\sigma_i \sqrt{2\pi}} e^{-(A_i - \mu_i)^2 / 2\sigma_i^2}$$

where μ_i is the mean, σ_i is the standard deviation, and σ_i^2 is the variance of attribute A_i over the training patterns of class C_k .

If we do not want to assume class conditional independence of the attributes, we use the Bayes classifier, not Naive Bayes. Then, adapting Eqn (5.1) to continuous attribute values, we maximize $P(C_k)p(\bar{A}|C_k)$. For that, we need, from the training set, the distribution of the attribute array $\bar{A} = A_1, A_2, \dots, A_M$ over class C_k . In practice, we often assume a multivariate normal distribution, hence we have

$$p(\bar{A}|C_k) = \frac{1}{\sqrt{|\Sigma|}(2\pi)^M} e^{[-(\bar{A} - \bar{\mu})^t \Sigma^{-1} (\bar{A} - \bar{\mu})]/2}$$

Let us now use Eqn (5.2) of the Naive Bayes classifier and do the same two classifications to check whether the results are identical. For the pattern whose attribute $A_1 = 1$:

$$P(C_1) \times P[(A_1 = 1)|C_1] = 0.3 \times 0.8 = 0.24$$

$$P(C_2) \times P[(A_1 = 1)|C_2] = 0.5 \times 0.4 = 0.20$$

$$P(C_3) \times P[(A_1 = 1)|C_3] = 0.2 \times 0.3 = 0.06$$

Since the value 0.24 obtained for class C_1 is the largest, the pattern is classified into class C_1 . This is the same as the classification done above by the linear classifier. Now let us classify the pattern whose attribute $A_1 = 0$:

$$P(C_1) \times P[(A_1 = 0)|C_1] = 0.3 \times 0.2 = 0.06$$

$$P(C_2) \times P[(A_1 = 0)|C_2] = 0.5 \times 0.6 = 0.30$$

$$P(C_3) \times P[(A_1 = 0)|C_3] = 0.2 \times 0.7 = 0.14$$

Since the value 0.30 obtained for class C_2 is the largest, the pattern is classified into class C_2 . This is the same as the classification done above by the linear classifier. So for both classifications, Eqn (5.2) and the linear classifier give identical results.

5.6 Continuous Attribute Values

Readers may be familiar with the terms *mean*, *variance*, *standard deviation*, and *covariance*. Nonetheless, Fig. 5.3 reviews these terms briefly.

In Section 5.2, we learnt how to estimate conditional probabilities when attribute values are discrete. Attribute values can, however, be continuous, for example, the height and weight of a person. In this section, we will discuss how to estimate conditional probabilities when attribute values are continuous.

Adapting Eqn (5.2) to Naive Bayes with continuous attribute values, we maximize $P(C_k) \prod_{i=1}^M p(A_i|C_k)$ for $1 \leq k \leq m$, the classes being C_1 to C_m . We need to obtain, from the training set, the distribution of attribute A_i over class C_k . In practice, we often assume the attribute to have a Normal

set, suppose we have estimated the following probabilities:

$$P(C_1) = 0.3$$

$$P[(A_1 = 1)|C_1] = 0.8 = a_{11}$$

$$P[(A_1 = 0)|C_1] = 0.2 = 1 - a_{11}$$

$$P(C_2) = 0.5$$

$$P[(A_1 = 1)|C_2] = 0.4 = a_{21}$$

$$P[(A_1 = 0)|C_2] = 0.6 = 1 - a_{21}$$

$$P(C_3) = 0.2$$

$$P[(A_1 = 1) | C_3] = 0.3 = a_{31}$$

$$P[(A_1 = 0) | C_3] = 0.7 = 1 - a_{31}$$

Let us classify a pattern whose attribute $A_1 = 1$. We need to evaluate the discriminant function as given in Eqn (5.11) for each of the three classes.

$$g_1(\bar{A}) = \log P(C_1) + \log a_{11} + A_1 \log \frac{a_{11}}{1 - a_{11}} = -0.6198$$

$$g_2(\bar{A}) = \log P(C_2) + \log a_{21} + A_1 \log \frac{a_{21}}{1 - a_{21}} = -0.6989$$

$$g_3(\bar{A}) = \log P(C_3) + \log a_{31} + A_1 \log \frac{a_{31}}{1 - a_{31}} = -1.2219$$

Since the value of $g_1(\bar{A})$ is the largest, the pattern with $A_1 = 1$ is classified into class C_1 . Now let us classify a pattern whose attribute $A_1 = 0$. We need to evaluate the discriminant function for each of the three classes.

$$g_1(\bar{A}) = \log P(C_1) + \log a_{11} + A_1 \log \frac{a_{11}}{1 - a_{11}} = -1.2219$$

$$g_2(\bar{A}) = \log P(C_2) + \log a_{21} + A_1 \log \frac{a_{21}}{1 - a_{21}} = -0.5228$$

$$g_3(\bar{A}) = \log P(C_3) + \log a_{31} + A_1 \log \frac{a_{31}}{1 - a_{31}} = -0.8539$$

Since the value of $g_2(\bar{A})$ is the largest, the pattern with $A_1 = 0$ is classified into class C_2 .

Substituting the right-hand side of the above equation for $\log P(A_i|C_k)$ in Eqn (5.10), we get

$$g_k(\bar{A}) = \log P(C_k) + \sum_{i=1}^M \log(1 - a_{ki}) + \sum_{i=1}^M A_i \log \frac{a_{ki}}{1 - a_{ki}} \quad (5.11)$$

This is a linear equation in A_i . The equation can be viewed as

$$g_k(\bar{A}) = w_{k0} + \sum_{i=1}^M w_{ki} A_i$$

where

$$w_{k0} = \log P(C_k) + \sum_{i=1}^M \log(1 - a_{ki})$$

and

$$w_{ki} = \log \frac{a_{ki}}{1 - a_{ki}}$$

for $0 < a_{ki} < 1$, such that $1 \leq k \leq m$ and $1 \leq i \leq M$. The w 's are viewed as *weights*, and $g_k(\bar{A})$ is the linear *discriminant function* for class C_k . Given the attribute array $\bar{A}$ of a pattern to be classified, the value of the discriminant function for each class is calculated, and then the pattern is assigned to the class with the discriminant function having the largest value. If each pattern is represented as a point in the M -dimensional A_1 - A_2 -...- A_M coordinate space, then each linear discriminant function represents a hyperplane that separates one class from another: the classes are *linearly separable*. A classifier that uses linear discriminant functions is said to be a *linear classifier*. Accordingly, a Naive Bayes classifier applied to patterns with binary attributes can be formulated as a linear classifier (more on linear classifiers is given in Sections 6.5, 7.2, and 7.3, and in Chapter 8).

Consider an example of three classes C_1 , C_2 , and C_3 . The attribute array $\bar{A}$ contains a single binary attribute A_1 . Therefore, $M = 1$. From a training

suppose we have estimated from the training set the prior probability $P(C_k)$ and the conditional probability $P(A_i|C_k)$, for $1 \leq k \leq m$ and $1 \leq i \leq M$, as described in Section 5.2. Let us define

$$a_{ki} = P[(A_i = 1)|C_k] \quad (5.8)$$

Since the binary attribute A_i can have a value either 0 or 1, we will have $P[(A_i = 1)|C_k] + P[(A_i = 0)|C_k] = 1$. Therefore, $P[(A_i = 0)|C_k] = 1 - P[(A_i = 1)|C_k]$. Accordingly, from Eqn (5.8), we can write

$$P[(A_i = 0)|C_k] = 1 - a_{ki} \quad (5.9)$$

To classify a pattern with attribute array $\bar{A}$, by the Naive Bayes classifier, we will be maximizing Eqn (5.2), that is, we will be maximizing

$$P(C_k) \prod_{i=1}^M P(A_i|C_k)$$

Since logs are monotonic, we can alternatively maximize

$$\log P(C_k) + \sum_{i=1}^M \log P(A_i|C_k)$$

We will evaluate the above equation for classes $C_1, C_2, \dots, C_m$ to find out for which class its value is the greatest, so that we can classify the pattern in that class. Let us define a function $g_k(\bar{A})$ such that

$$g_k(\bar{A}) = \log P(C_k) + \sum_{i=1}^M \log P(A_i|C_k) \quad (5.10)$$

The second term in the above equation

$$\begin{aligned} \log P(A_i|C_k) &= A_i \log a_{ki} + (1 - A_i) \log(1 - a_{ki}) \\ &\text{identity holds for possible values 0 and 1 of } A_i \\ &\text{with } a_{ki} \text{ as defined in Eqns (5.8) and (5.9)} \end{aligned}$$

$$= \log(1 - a_{ki}) + A_i [\log a_{ki} - \log(1 - a_{ki})]$$

$$= \log(1 - a_{ki}) + A_i \log \frac{a_{ki}}{1 - a_{ki}}$$

In Section 5.3, we have not considered risk when we classify recall pattern R3 of the professor-student training set given in Fig. 1.2 by maximizing Eqn (5.2).

Suppose we need to reclassify the pattern by minimizing risk, where $\lambda(\text{professor}) = 2$ and $\lambda(\text{student}) = 1$. In other words, the loss in misclassifying a professor is twice the loss in misclassifying a student. The attribute values for the pattern are HABIT = gabby, EATS = roasted, and FOOTWEAR = clogs. Using the Bayesian probability estimates from Fig. 5.1, and the Naive Bayes classifier of Eqn (5.7), the value of

$$\begin{aligned} &\lambda(\text{professor}) \times P(\text{professor}) \times P[(\text{HABIT} = \text{gabby}) | \text{professor}] \\ &\times P[(\text{EATS} = \text{roasted}) | \text{professor}] \times P[(\text{FOOTWEAR} = \text{clogs}) | \text{professor}] \end{aligned}$$

is $2 \times 2/5 \times 3/5 \times 1/3 \times 1/5 = 0.032$, and the value of

$$\begin{aligned} &\lambda(\text{student}) \times P(\text{student}) \times P[(\text{HABIT} = \text{gabby}) | \text{student}] \\ &\times P[(\text{EATS} = \text{roasted}) | \text{student}] \times P[(\text{FOOTWEAR} = \text{clogs}) | \text{student}] \end{aligned}$$

is $1 \times 3/5 \times 4/7 \times 1/8 \times 4/7 = 0.0245$.

Since the value 0.032 obtained for professor is more than the 0.0245 obtained for student, recall pattern R3 is classified as professor. This contrasts with the classification in Section 5.3, where, by not considering risk, recall pattern R3 is classified as student.

For the rest of the discussion in this chapter, it will be assumed that $\lambda(C_k) = 1$, for $1 \leq k \leq m$, that is, the loss in any misclassification is 1. Under this assumption, Eqn (5.7) simplifies to Eqn (5.2). It can be said that Eqn (5.2) is a special case of Eqn (5.7).

5.5 Naive Bayes with Binary Attributes

A *binary* attribute has two possible values; for example, attribute HABIT in the professor-student patterns given in Fig. 1.1 has two possible values: gabby and quiet. We could as well have had HABIT's possible values being 0 and 1. In general, any binary attribute can be coded so that the possible values are 0 and 1.

Consider a training set of patterns whose every attribute A_1 to A_M is binary, each attribute's possible values being 0 and 1. For the m classes C_1 to C_m ,

5.4 Minimizing Risk

Some misclassifications can cause more loss than others: classifying a cancerous tumour as benign may lead to premature death; classifying a benign tumour as cancerous, to unnecessary therapy. The former is considered to cause more loss than the latter.

Let us assume there is no loss when a pattern is classified correctly, but there is a loss when the pattern is misclassified. For the m classes C_1 to C_m , let $\lambda(C_i)$ be the loss in misclassifying a pattern of class C_i , where $1 \leq i \leq m$. This means we are assuming that the loss is the same regardless of which class the pattern is misclassified into. Suppose there is a pattern with attribute array $\bar{A}$, and we need to classify it. The expected loss (known as the *risk*) in classifying the pattern into class C_k is

$$\sum_{i \neq k}^m \lambda(C_i) P(C_i | \bar{A})$$

There will be no loss if the pattern actually belongs to class C_k , hence we have $i \neq k$ with the summation sign. The above equation can be rewritten as

$$\sum_{i=1}^m \lambda(C_i) P(C_i | \bar{A}) - \lambda(C_k) P(C_k | \bar{A})$$

We want to minimize the risk in classifying, therefore we minimize the above equation. The first term of the equation is a constant. Accordingly, we minimize the above equation by maximizing

$$\lambda(C_k) P(C_k | \bar{A})$$

In other words, to minimize the risk, we classify the pattern into that class C_k for which the above equation has the highest value. Following the same reasoning as that used to derive Eqn (5.2), we can say that, to minimize risk in the Naive Bayes classifier, we maximize

$$\lambda(C_k) P(C_k) \prod_{i=1}^M P(A_i | C_k) \quad (5.7)$$

is, for $1 \leq i \leq m$; (b) the program uses the probability estimates it had made in the training phase to classify a given training or recall pattern. Use this program to repeat the first four exercises. It is shown in Section 5.4 that when the loss of every misclassification is 1, Eqn (5.7) becomes Eqn (5.2).

6. Use the program you wrote in the fifth exercise above to train a Naive Bayes classifier on the training set of computer science departments described in the tenth exercise of Chapter 2. Report your results of classifying the patterns of the recall set.
7. Extend the program you wrote in the fifth exercise above so that it can be used as a linear classifier when all attributes are binary.
8. In Section 3.1, you read how decision trees are adapted to handle missing attribute values. Can the Bayes procedure be adapted to handle missing attribute values? If not, why? If yes, describe the adaptations required. Remember, there may be missing attribute values in the training set. So, changes may be required in the estimation of probabilities from the training set. When there are missing attribute values in a recall pattern, changes may be required in the classification of the pattern.